Practice Set A

Class 12 Mathematics, Chapter 11: Three Dimensional Geometry

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, CBSE 2024 — 1 mark

Direction ratios of a vector parallel to the line $\frac{x-1}{2} = -y = \frac{2z+1}{6}$ are:

(A) $2, -1, 6$ **(B)** $2, 1, 6$ **(C)** $2, 1, 3$ **(D)** $2, -1, 3$

2 MCQ, CBSE 2024 — 1 mark

If a line makes an angle of 30° with the positive direction of x-axis, 120° with the positive direction of y-axis, then the angle which it makes with the positive direction of z-axis is:

(A) 90° **(B)** 120° **(C)** 60° **(D)** 0°

3 MCQ, CBSE 2024 — 1 mark

The Cartesian equation of a line passing through the point with position vector $\vec{a} = \hat{i} - \hat{j}$ and parallel to the line $\vec{r} = \hat{i} + \hat{k} + \mu(2\hat{i} - \hat{j})$ is:

(A) $\frac{x-2}{1} = \frac{y+1}{0} = \frac{z}{1}$ **(B)** $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{0}$

(C) $\frac{x+1}{2} = \frac{y+1}{-1} = \frac{z}{0}$ **(D)** $\frac{x-1}{2} = \frac{y}{-1} = \frac{z-1}{0}$

4 MCQ, CBSE 2023 — 1 mark

The angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$ is:

(A) 0° **(B)** 30° **(C)** 45° **(D)** 90°

5 MCQ, CBSE 2024 — 1 mark

The lines $\frac{1-x}{2} = \frac{y-1}{3} = \frac{z}{1}$ and $\frac{2x-3}{2p} = \frac{y}{-1} = \frac{z-4}{7}$ are perpendicular to each other for p equal to:

(A) $-\frac{1}{2}$ **(B)** $\frac{1}{2}$ **(C)** 2 **(D)** 3

Q No.	Question	Marks
6	Assertion-Reason, CBSE 2023 **Assertion (A):** If a line makes angles α, β, γ with the positive directions of the coordinate axes, then $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$. **Reason (R):** The sum of squares of the direction cosines of a line is 1. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, PYQ-style Find the vector and Cartesian equations of the line passing through the points $(3, 4, -7)$ and $(1, -1, 6)$.	2 marks

Level 2

16 marks

8	Short answer, PYQ-style Show that the line through the points $(1, -1, 2)$ and $(3, 4, -2)$ is perpendicular to the line through the points $(0, 3, 2)$ and $(3, 5, 6)$.	2 marks
9	Short answer, CBSE SQP 2024-25 Find the shortest distance between the lines l_1 and l_2 whose vector equations are $\vec{r} = (-\hat{i} - \hat{j} - \hat{k}) + \lambda(7\hat{i} - 6\hat{j} + \hat{k})$ and $\vec{r} = (3\hat{i} + 5\hat{j} + 7\hat{k}) + \mu(\hat{i} - 2\hat{j} + \hat{k})$, where λ and μ are parameters.	3 marks
10	Short answer, CBSE 2024 Find the distance between the line $\frac{x}{2} = \frac{2y-6}{4} = \frac{1-z}{-1}$ and another line parallel to it passing through the point $(4, 0, -5)$.	3 marks
11	Short answer, PYQ-style Find the angle between the lines $\vec{r} = 3\hat{i} + \hat{j} - 2\hat{k} + \lambda(\hat{i} - \hat{j} - 2\hat{k})$ and $\vec{r} = 2\hat{i} - \hat{j} - 56\hat{k} + \mu(3\hat{i} - 5\hat{j} - 4\hat{k})$.	2 marks
12	Short answer, PYQ-style Find the value of k so that the lines $x = -y = kz$ and $x - 2 = 2y + 1 = -z + 1$ are perpendicular to each other.	2 marks
13	Short answer, CBSE 2023 If the equation of a line is $x = ay + b$, $z = cy + d$, then find the direction ratios of the line and a point on the line.	2 marks
14	Short answer, CBSE SQP 2026-27 A line passing through the points $(2, 6, 5)$ and $(3, p, 7)$ is parallel to the line joining the points $(-4, 3, 4)$ and $(-1, 0, q)$. Find the value of $\frac{q}{p}$.	2 marks

Q No.	Question	Marks

Level 3 — 27 marks

15 Long answer, CBSE SQP 2026-27 — 5 marks

Find the foot of the perpendicular drawn from the point $(2, 1, 4)$ on the line $x - 2 = \frac{2y + 3}{4} = 2 - z$. Also find the equation of the line passing through this point and the point $(2, 1, 4)$.

16 Long answer, CBSE SQP 2024-25 — 5 marks

Find the image of the point $(1, 2, 1)$ with respect to the line $\frac{x - 3}{1} = \frac{y + 1}{2} = \frac{z - 1}{3}$. Also find the equation of the line joining the given point and its image.

17 Long answer, CBSE 2025 — 5 marks

Find the distance of the point $(-1, -5, -10)$ from the point of intersection of the lines $\frac{x - 1}{2} = \frac{y - 2}{3} = \frac{z - 3}{4}$ and $\frac{x - 4}{5} = \frac{y - 1}{2} = z$.

18 Long answer, CBSE SQP 2025-26 — 5 marks

The two lines $\frac{x - 1}{3} = -y,\ z + 1 = 0$ and $\frac{-x}{2} = \frac{y + 1}{2} = z + 2$ intersect at a point whose y-coordinate is 1. Find the coordinates of their point of intersection. Find the vector equation of a line perpendicular to both the given lines and passing through this point of intersection.

19 Short answer, CBSE SQP 2024-25 — 3 marks

Find the vector and the Cartesian equations of the line that passes through $(-1, 2, 7)$ and is perpendicular to the lines $\vec{r} = 2\hat{i} + \hat{j} - 3\hat{k} + \lambda(\hat{i} + 2\hat{j} + 5\hat{k})$ and $\vec{r} = 3\hat{i} + 3\hat{j} - 7\hat{k} + \mu(3\hat{i} - 2\hat{j} + 5\hat{k})$.

20 Case study, CBSE SQP 2026-27 — 4 marks

Case Study: Two security sensors in a multi-storey mall emit laser beams for movement detection. Three of these beams travel along the following lines: $l_1 : \vec{r} = (\lambda + 1)\hat{i} + (3\lambda + 2)\hat{j} + (3 - \lambda)\hat{k}$, $l_2 : \vec{r} = 2(\mu + 1)\hat{i} + (3\mu - 1)\hat{j} + (\mu + 1)\hat{k}$ and $l_3 : \vec{r} = (\gamma + 3)\hat{i} + (3\gamma - 1)\hat{j} + (1 - \gamma)\hat{k}$. Based on the given information, answer the following questions.

(i) Show that the lines l_1 and l_2 are not parallel to each other. — 1 mark

(ii) Show that the lines l_1 and l_3 are parallel to each other. — 1 mark

(iii) Find the shortest distance between the lines l_1 and l_2. — 2 marks

Answer key

Chapter 11: Three Dimensional Geometry, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(D) $2, -1, 3$
2	(A) 90°
3	(B) $\dfrac{x-1}{2} = \dfrac{y+1}{-1} = \dfrac{z}{0}$
4	(D) 90°
5	(C) 2
6	(A) Both true; R is the correct explanation of A.
7	$\vec{r} = (3\hat{i} + 4\hat{j} - 7\hat{k}) + \lambda(-2\hat{i} - 5\hat{j} + 13\hat{k})$; $\dfrac{x-3}{-2} = \dfrac{y-4}{-5} = \dfrac{z+7}{13}$

Level 2

8	$a_1a_2 + b_1b_2 + c_1c_2 = 6 + 10 - 16 = 0$, so the lines are perpendicular.
9	$2\sqrt{29}$ units
10	$\dfrac{\sqrt{533}}{3}$ units
11	$\theta = \cos^{-1}\left(\dfrac{8\sqrt{3}}{15}\right)$
12	$k = 2$
13	DRs $a, 1, c$; a point on the line is $(b, 0, d)$.
14	$\dfrac{q}{p} = 2$

Level 3

15	Foot $P\left(\dfrac{5}{2}, -\dfrac{1}{2}, \dfrac{3}{2}\right)$; line $\dfrac{x-2}{1} = \dfrac{y-1}{-3} = \dfrac{z-4}{-5}$
16	Image $Q\left(\dfrac{39}{7}, -\dfrac{20}{7}, \dfrac{19}{7}\right)$; line PQ: $\dfrac{x-1}{16} = \dfrac{y-2}{-17} = \dfrac{z-1}{6}$
17	Point of intersection $(-1, -1, -1)$; distance $= \sqrt{97}$ units
18	Point $(-2, 1, -1)$; $\vec{r} = (-2\hat{i} + \hat{j} - \hat{k}) + \lambda(\hat{i} + 3\hat{j} - 4\hat{k})$
19	$\vec{r} = (-\hat{i} + 2\hat{j} + 7\hat{k}) + \lambda(10\hat{i} + 5\hat{j} - 4\hat{k})$; $\dfrac{x+1}{10} = \dfrac{y-2}{5} = \dfrac{z-7}{-4}$
20	(i) DRs not proportional; (ii) same direction $\hat{i} + 3\hat{j} - \hat{k}$; (iii) $\dfrac{7\sqrt{6}}{6}$ units